

# Siddharth Singh

## Software Engineer



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<https://github.com/demiqod-ares>

## EXPERIENCE

### Software Engineer 2 - JP Morgan Chase

~ 04/2025 - Working, Mumbai - CDAI - Consumer & Community Banking Technology - Consumer Card Technology, Card Data Product

#### Key Achievements:

- Led engineering **delivery for Consumer Card data product platforms**, converting product and business requirements into scalable cloud-native services, data workflows, and deployment automation that improved delivery predictability, operational resilience, and stakeholder confidence.
- Built and maintained **AWS cloud infrastructure using Terraform**, enabling repeatable environment provisioning, stronger configuration control, reduced environment drift, and faster onboarding of new application components across development and test environments.
- Designed and operated **Apache Airflow and PySpark based data workflows** supporting card data processing, data product enablement, and downstream consumption needs, improving pipeline reliability, data freshness, and SLA adherence.
- Contributed to **data marketplace enablement** by supporting reusable data product patterns, improving discoverability, controlled access, and consumption of curated Consumer Card datasets by downstream teams.
- Implemented **Release Orchestrator-based deployment automation** to coordinate multi-component releases across application/service IDs, enabling controlled Stop/Deploy/Start sequencing and reducing manual release execution effort by 70%.
- Standardized deployment patterns using **Spinnaker, Jules, AAAS/AIM-integrated workflows, and Release Orchestrator**, improving release consistency, reducing coordination overhead, and increasing deployment confidence across environments.
- Advanced adoption of **TrueCD testing and release automation practices**, helping shift quality left, strengthen production readiness, reduce late-cycle defects, and support faster delivery without compromising controls or software quality.
- Improved observability and operational support using **Dynatrace and Splunk**, creating dashboards, alerts, and investigation workflows that reduced mean time to detect and recover from incidents by 82%.
- Partnered with product owners, QA, platform engineering, and application teams to manage dependencies, refine delivery scope, prioritize backlog items, and **deliver production-ready capabilities** aligned with Consumer Card business priorities.
- Strengthened engineering governance** by improving CI/CD hygiene, deployment readiness checks, runbooks, and operational playbooks, reducing recurring production support effort and improving auditability of release activities.

### Software Engineer 1 - JP Morgan Chase

~ 01/2024 - Working, Mumbai - RDS - Commercial & Investment Bank Technology - Payments Technology, Global Liquidity, Cash Management / Liquidity Services

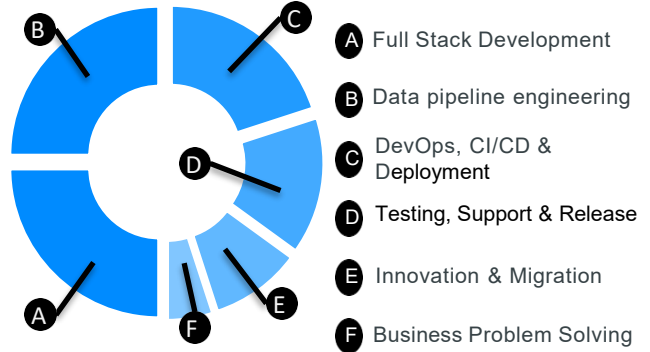
#### Key Achievements:

- Developed and supported backend services** (Java/Spring Boot) for liquidity and cash management platforms, contributing to reliable payment and liquidity processing capabilities achieving 99.99% uptime.
- Built and enhanced service integrations** supporting Global Liquidity & Cash Management workflows, improving transaction processing reliability, reducing operational exceptions, supporting business-critical liquidity services.
- Automated batch schedules** and dependency-driven processing using Autosys, improving job execution reliability, reducing manual interventions, and helping meet time-sensitive operational SLAs.
- Supported CI/CD and deployment processes** through Jules, improving deployment repeatability, reducing release coordination effort, and enabling safer application changes across environments.
- Investigated production issues** across services, jobs, and integration points, delivering root-cause fixes and operational improvements that reduced repeat incidents and improved platform stability.

## SUMMARY

Software Development Engineer with experience building cloud-native data platforms, deployment automation, and enterprise-grade backend services across Consumer Card, Liquidity Technology and Supply Chain. Skilled in AWS, Terraform, Apache Airflow, PySpark, Java, Spring Boot, CI/CD, observability, and production resilience.

## MY COMPETENCY



## SKILLS

**Server-side:** NodeJS, NestJS, Jest, Java, Spring Boot, JUnit, Hibernate, Postman, Microservices

**Client-Side:** TypeScript, JavaScript, Angular, React, Redux, Micro-frontend

**Cloud & Containers:** AWS (Lambda, S3, Glue, SQS, Cloudwatch), Azure, Terraform, Docker, Kubernetes, Docker swarm

**Data Engineering:** Apache airflow, Pyspark

**Observability & Monitoring:** Splunk, Dynatrace

**Databases, Cache:** MySQL, Oracle, Postgres, Liquibase, CouchDB, MongoDB, Cassandra, Kafka, Zookeeper, IBM MQ, Rabbit MQ, Gemfire, Redis

**CI/CD & Automation:** Jenkins, Spinnaker, Concourse, Shell scripting, Autosys, n8n.io

**Management Frameworks:** Scaled Agile (SAFE), LEAN, Agile, Scrum, Fast Works

## AWARDS



Manager's Choice Award @IBM



Gold Medal 2020 @NIT Kurukshetra



Gold Medal 2015 @NPTI Badarpur

- **Collaborated** with business analysts, operations users, QA, and application teams to clarify requirements, validate implementation behavior, and ensure production readiness for liquidity platform enhancements.
- **Tech Stack Modernization & Security Overhaul:** Eliminated 100% of SSAP/SonarQube vulnerabilities post Java, Springboot and React version upgrade, achieving 75% Sonar coverage
- **Kafka Infrastructure Leadership:** Classified 50+ Kafka topics enabling role-based access for 15+ downstream systems.
- **Developed and enhanced React and Angular based micro frontends**, delivering reusable UI components, improving user experience, and supporting scalable UI capabilities across business-critical institutional client workflows.

## Application Developer (Promotions 6B<6A<6G) - IBM India Pvt Ltd

~ 04/2021 – 10/2023, Bangalore - Vinturas - Private Blockchain (Hyperledger Fabric) based supply chain ecosystem

### Key Achievements:

- **Cost-Efficient Architecture:** Slashed IKS logging costs by 35% and improved incident response time by 50% via Rancher/Grafana monitoring and automated n8n.io workflows, saving 20 hours/week in manual tasks.
- **CI/CD Innovation:** Streamlined Concourse pipelines to accelerate deployment cycles by 40%, enabling zero-downtime updates for 10+ Dockerized microservices.
- **Blockchain Advancements:** Built 5+ EVM D-apps (Solidity, Truffle) with MetaMask integration, reducing transaction latency by 60% and scaling user onboarding to 200+/month.
- **Client-Centric Development:** Translated 50+ stakeholder requirements into JIRA user stories, reducing sprint backlogs by 30% and aligning cross-functional teams.
- **Microservice Excellence:** Containerized 15+ microservices using Docker, achieving 99.8% test coverage through rigorous unit, regression, and E2E testing.

## Researcher

### NIT Kurukshetra

~ 07/2018 - 08/2020, Haryana

- Conducted in-depth research for my thesis focused on "Advanced Optimization Techniques for Engineering Applications." Implemented modeling and solutions for intricate challenges encompassing Multi-period Dynamic Economic Dispatch, Unit Commitment, and Multi-period Optimal DC/AC Power Flow.
- Explored areas including Energy Storage System integration, Power System observability enhancement via PMU technology, and Transmission Network operation and planning.
- Employed various optimization methods such as Non-Linear Programming, Mixed Integer NLP, and Relaxed MINLP to integrate energy systems.
- Additionally, delved into learning state-of-the-art Meta-heuristics and Machine Learning Algorithms for enhanced problem-solving capabilities.

## EDUCATION

M.Tech (Power Electronics)

NIT Kurukshetra

~ 06/2018 - 06/2020

%

87 / 100

PG Diploma (Industrial Automation)

Prolific Systems & technology, Noida

~ 02/2015 - 07/2015

PG Diploma (Power, T&D)

NPTI Badarpur, New Delhi

~ 07/2014 - 02/2015

%

78 / 100

B.Tech (EEE)

IMS Engineering Collage, Ghaziabad

~ 07/2010 - 07/2014

%

78 / 100

## PUBLICATIONS

Open-Source Contribution : AI & ML

GitHub

<https://github.com/demigod-ares/>

OSC : Power System optimization

GitHub

<https://github.com/demigod-ares/Microgrid-Basics-GAMS-codes>

Network Constrained AC/DC

Economic Dispatch for DER based

Microgrid

IEEE- India International Conference

~ 03/2020

□ <https://ieeexplore.ieee.org/document/9112982>

Economic dispatch for multi-heat

electric energy source based

microgrid

IEEE- PIICON

~ 02/2020

□ <https://ieeexplore.ieee.org/document/9112911>

Steady and dynamic response of

multi-machine system with various

FACTS devices using PSAT

Journal of The Institution of Engineers

~ Under review

Comparison of microgrid with voltage

dependent load based upon DER and

BES availability

Renewable Energy, Elsevier Journal

~ Under review

Robust Optimization Framework for

Economic Dispatch of DER and BES

based Microgrid implementing a Time-

Varying Voltage- Dependent Load to

achieve Conservation Voltage

Reduction

Majlesi Journal of Electrical Engineering

~ Under review